

The Utility Problem

Capacity is important in a distribution grid because it determines the amount of electrical power that can be supplied to customers. It is determined by multiple attributes, for example, the size and rating of the conductors and transformers used in the feeder line, as well as by the loading conditions, including the number of customers connected to the feeder, the load density, and the expected maximum demand.

If the capacity of the distribution grid is not sufficient, it can result in the overloading of the system, leading to power quality issues and even power outages. This can impact the reliability and quality of service of the distribution grid for customers and may also result in equipment damage.

Therefore, it is important to have adequate capacity in a distribution grid to ensure that there is enough electrical power available to meet the current and future demand for electricity. This can be done by upgrading existing infrastructure, adding new feeders or laterals, or by optimizing the use of existing capacity through demand management and load balancing. On the other hand, it is not economically efficient to oversize the conductor cross-sections unnecessarily, as a higher cross-section is more expensive and unnecessarily higher investment costs would be spent.

The Data Solutions and Methods

In the North American grid at the low-voltage level, there is typically a three-phase feeder from the transformer, and then single-phase feeders (laterals) are routed directly to customers via taps from this feeder. The capacity must be calculated for both feeder and lateral.

Data required for the calculation:

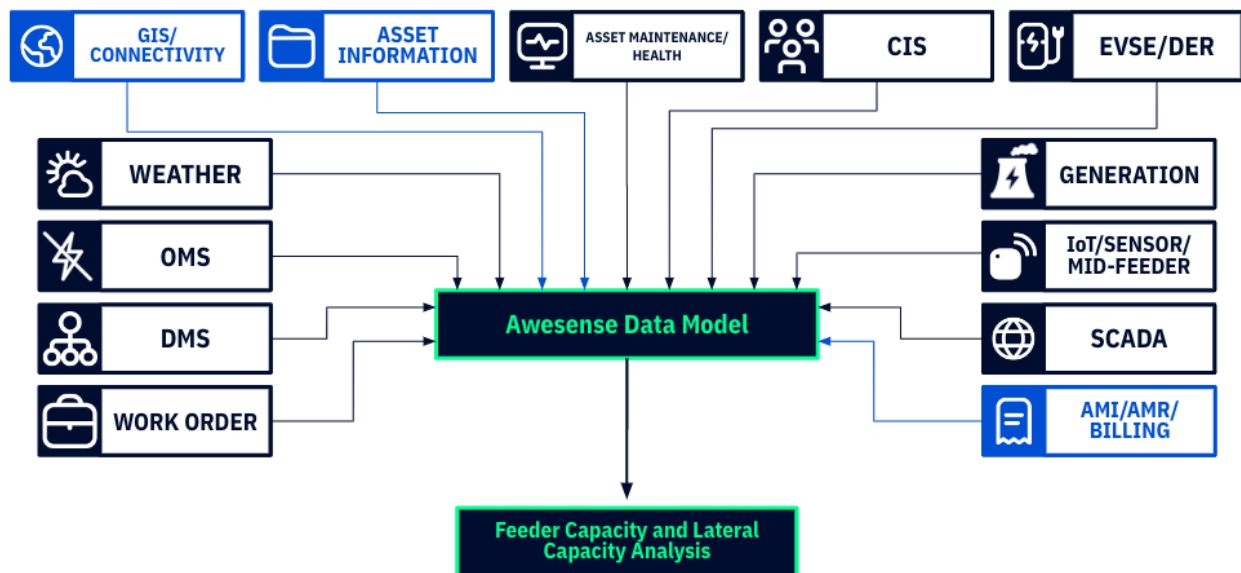
1. GIS data and tracing - GIS data allows to identify which sections belong to each feeder and which consumption points belong to the feeder and these sections.
2. Asset data - the maximum possible load capacity (from cross-section and conductor type) must be assigned to each feeder section. Each section may have a different cross-section and conductor type - typically, at the transformer, there will be a higher cross-section as it is necessary to supply power to all the consumption points at the feeder, and further away from the transformer, the cross-section may decrease as there is no longer a need to supply power to as many consumption points as at the beginning of the feeder - this is known as the economic cross-section.
3. Load on the feeder and its individual sections - the load on the feeder can be determined using time series data of the individual consumption points. These consumption points are summed for each section according to the topology.

Finally, the load of each section is compared along with the capacity of its conductor. This allows us to know the maximum load of each section and thus to know whether the section needs to be reinforced or whether it still has sufficient spare capacity.

Awesense Energy Transition Platform is ready for such analysis. By integrating a wide variety of data, all the necessary data is available in one place (the data is cleaned and synchronized) and is easy to access thanks to sophisticated functions for, for example, grid tracing and exports based on this tracing or time series data aggregation.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. A GitHub link to the Jupyter notebook can be provided upon request.